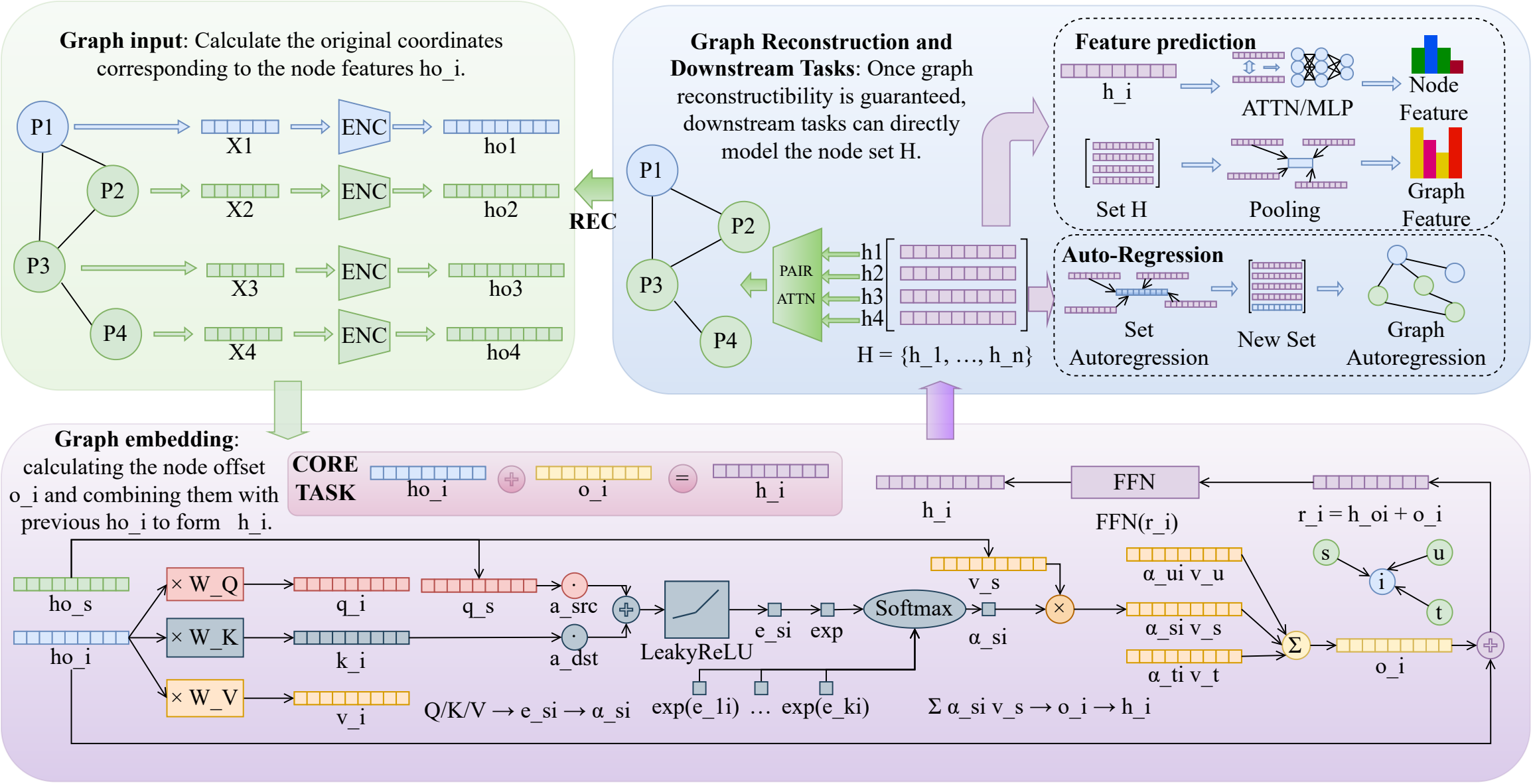

Graph input: Calculate the original coordinates corresponding to the node features ho_i.
P1
P2
P3
P4
X1
X2
X3
X4
ENC
ho1
ho2
ho3
ho4
REC
Graph Reconstruction and Downstream Tasks: Once graph reconstructibility is guaranteed, downstream tasks can directly model the node set H.
PAIR
ATTN
h1
h2
h3
h4
H = {h_1, …, h_n}
Feature prediction
h_i
ATTN/MLP
Node Feature
Set H
Pooling
Graph Feature
Auto-Regression
Set Autoregression
New Set
Graph Autoregression
Graph embedding: calculating the node offset o_i and combining them with previous ho_i to form h_i.
CORE TASK
ho_i
o_i
h_i
FFN
FFN(r_i)
r_i = h_oi + o_i
ho_s
ho_i
× W_Q
× W_K
× W_V
q_i
k_i
v_i
q_s
a_src
a_dst
LeakyReLU
e_si
exp
Softmax
α_si
v_s
α_ui v_u
α_si v_s
α_ti v_t
Σ
o_i
s
u
i
t
Q/K/V → e_si → α_si
exp(e_1i) … exp(e_ki)
Σ α_si v_s → o_i → h_i